

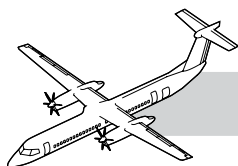
CHAPTER 57

WINGS

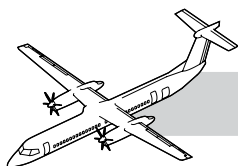
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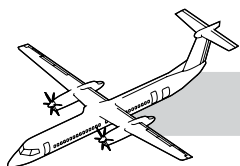
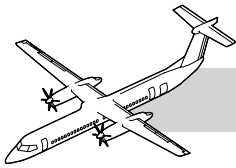


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CHAPTER 57 WINGS



57-00-00 WINGS GENERAL

INTRODUCTION

Refer to Figure 57-1. Center and Outer Wing Structures.

The main structure is of a box construction and, consists of machined front, and rear spars which are inter connected by chordwise ribs, lateral stringers, upper and lower bonded skin.

The wing is an all metal, fully cantilever unit mounted through the upper cabin of the fuselage, incorporating ailerons, spoilers and trailing edge flaps.

It is also used for fuel storage, including the collector and surge bays.

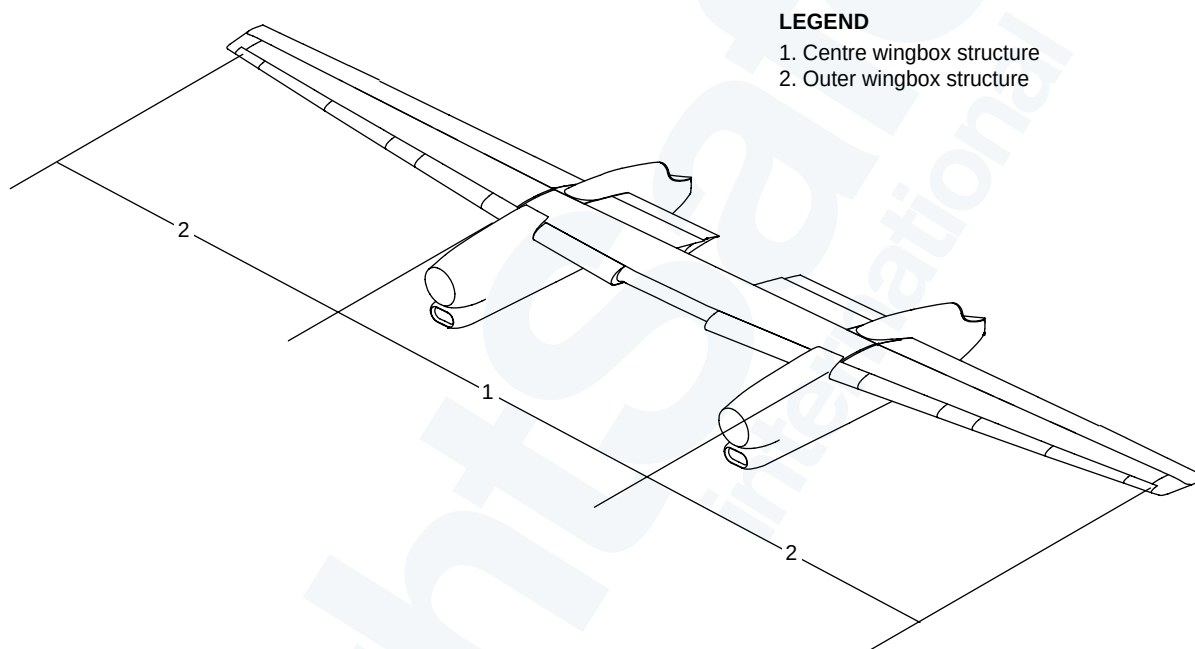
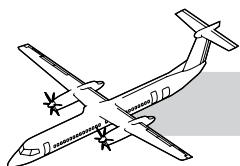
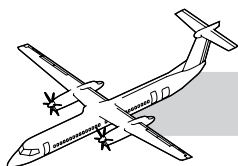


Figure 57-1. Center and Outer Wing Structures



The wing consists of the following seven main sections:

- Center wing
- Outer wing
- Wing tip
- Wing leading edge
- Flaps
- Ailerons
- Spoilers.

NOTES

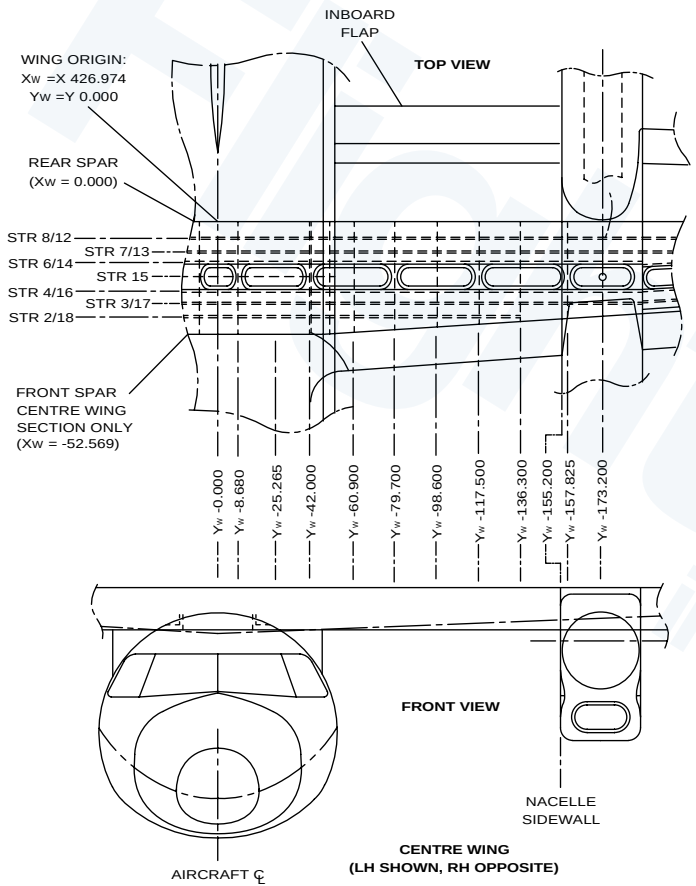
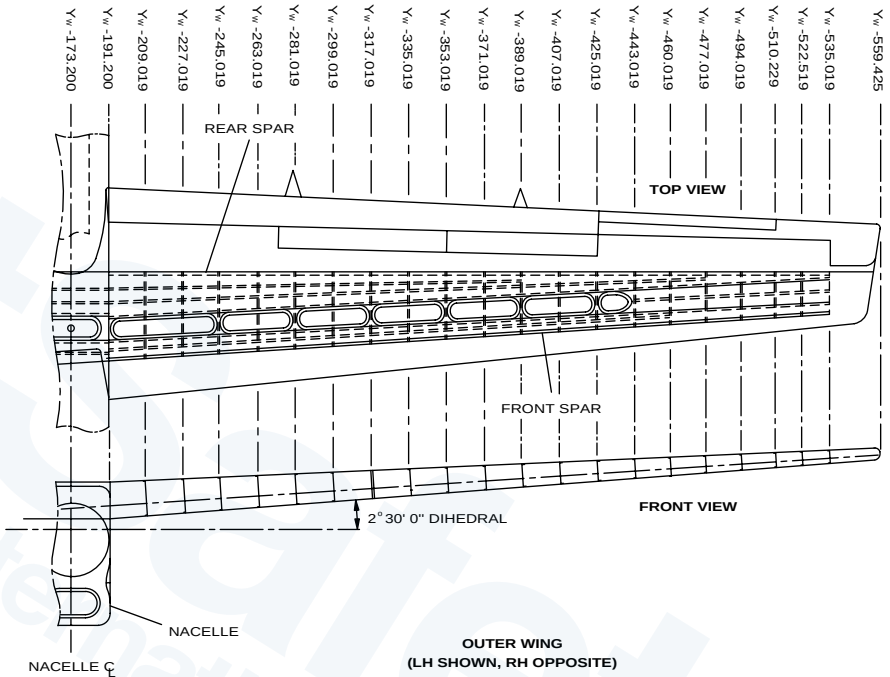
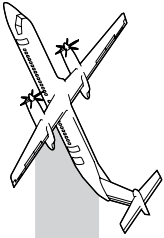
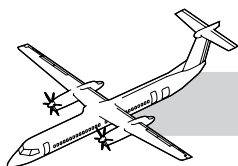


Figure 57-2. Aircraft Wings Station



GENERAL

Refer to Figure 57-2. Aircraft Wings Station.

Main Structure:

The wing box structure forms the integral fuel tanks and also provides the mounting structure for the nacelles. Integrally machined ladder plates located on the upper wing surface, provide access to the integral fuel tanks.

Auxiliary structure:

The auxiliary structure consists of the wing leading edges, flap and aileron shroud, spoilers and wing tips.

Leading Edge:

The leading edge is made of composite construction, using chordwise stiffeners. It can be removed to provide access to components and equipment mounted on the forward face of the front spar.

Flap Shroud:

The flap shroud, inboard and outboard, consists of separate sections riveted to the wing. Cavities are formed in the upper surface to provide flush mounting of the spoilers.

Aileron Shroud:

The aileron shroud is attached to the wing rear spar and it includes the support structure for the aileron control quadrant.

Wing Tip:

The wing tip is of composite construction and has a mounting bracket for the navigation light.

Flaps:

The wing flaps are positioned between stations Yw 49.09 and Yw 154.70 (Inner), and between Yw 191.419 and Yw 425.019 (Outer). The main box frame is made of aluminum alloy, using front and rear spars, chordwise ribs, spanwise stiffeners and bonded skin. The leading and trailing edges are made of composite material and use chordwise stiffeners for strength.

Ailerons:

The ailerons use aluminium alloy materials for spars, chordwise stiffeners and skin. The aileron has a geared tab constructed from an aluminium alloy.

Spoilers:

The spoilers are of composite construction with aluminium skin. All spoilers are flush mounted on the upper surface of the flap shroud, behind the wing rear spar.

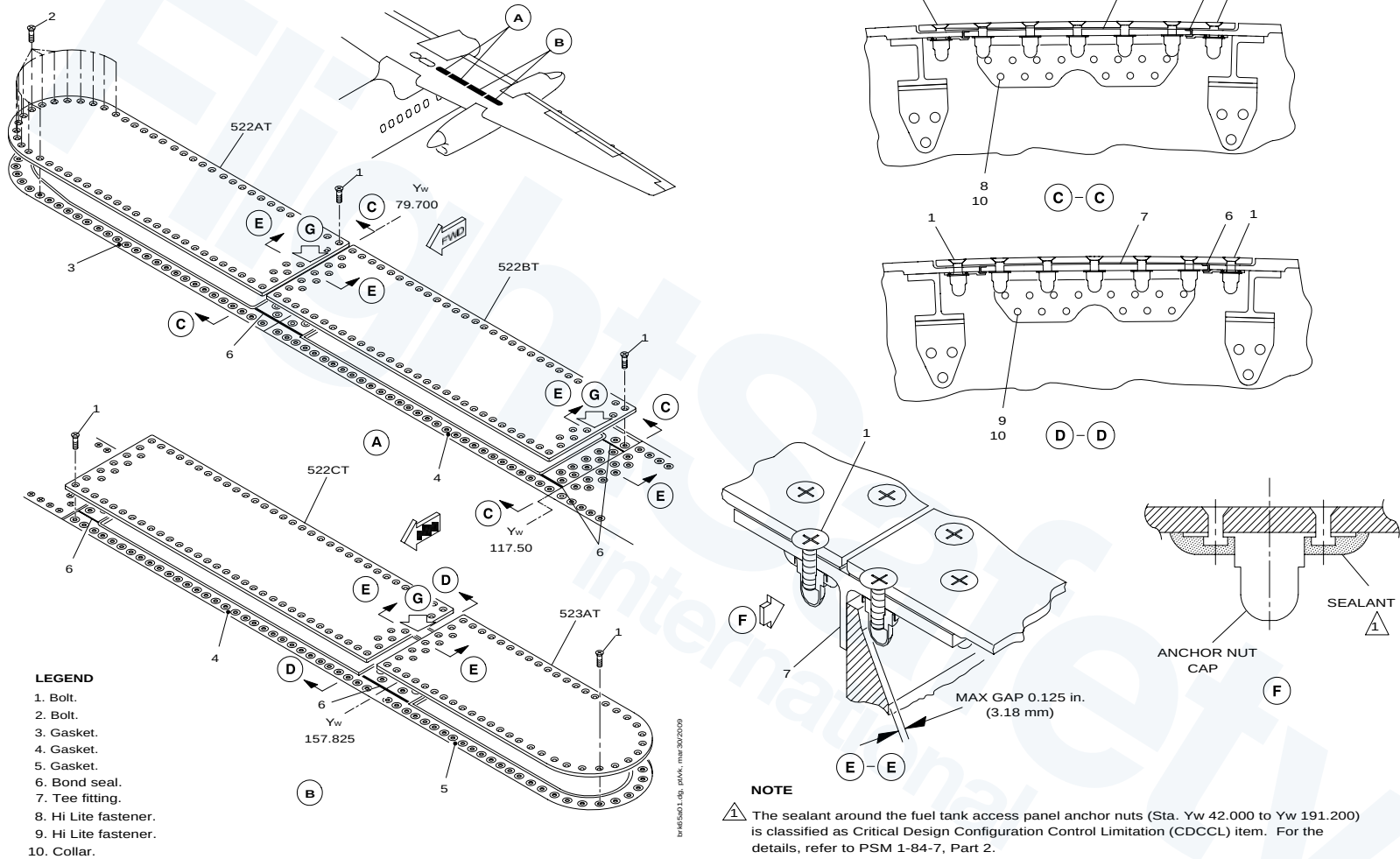
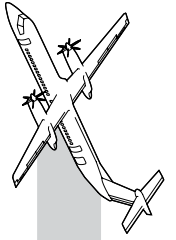
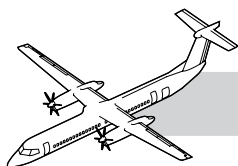


Figure 57-3. Center Wing Access Panels, Wet Bay



57-10-00 CENTER WING

INTRODUCTION

The center wing stations are from Yw 192.142 to Yw -192.142.

GENERAL

The center wing is an all metal unit mounted through the upper cabin of the fuselage and incorporates the inboard flaps.

The center wing box structure forms part of the integral fuel tanks and also provides the mounting structure for the nacelles.

Structural ladder plates located on the upper wing surface, give access to the center wing integral fuel tanks.

The center wing structure has the following components:

- Stringers
- Spar
- Ribs
- Wing to fuselage fittings
- Center wing box skin
- Access panels.

COMPONENT DESCRIPTION

Center Wing Access Panels, Wet Bay

Refer to Figure 57-3. Center Wing Access Panels, Wet Bay.

The wet bay center wing access panels are installed with screws.

The panels are made of machined aluminium alloy plate 7150-T7751. Gaskets with metallic washers are used in this location.

Before you remove a wing access panel in the wet bay, you must perform the following procedure:

- Make sure that the aircraft is grounded
- On the DC CONTROL panel, make sure that all battery switches are in the OFF position
- Remove all the fuel from the tanks
- Jack the aircraft using the front fuselage and the wing jacks
- Lift the engine to remove the wing twist.

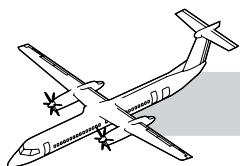
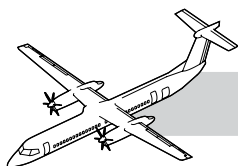


Figure 57-4. Load Cell Kit - Procure Locally



Figure 57-5. Load Cell and Hoist



Special Tooling Required

Refer to:

- Figure 57-4. Load Cell Kit - Procure Locally.
- Figure 57-5. Load Cell and Hoist.
- Figure 57-6. Hoist - Nacelle Mounted Engine.

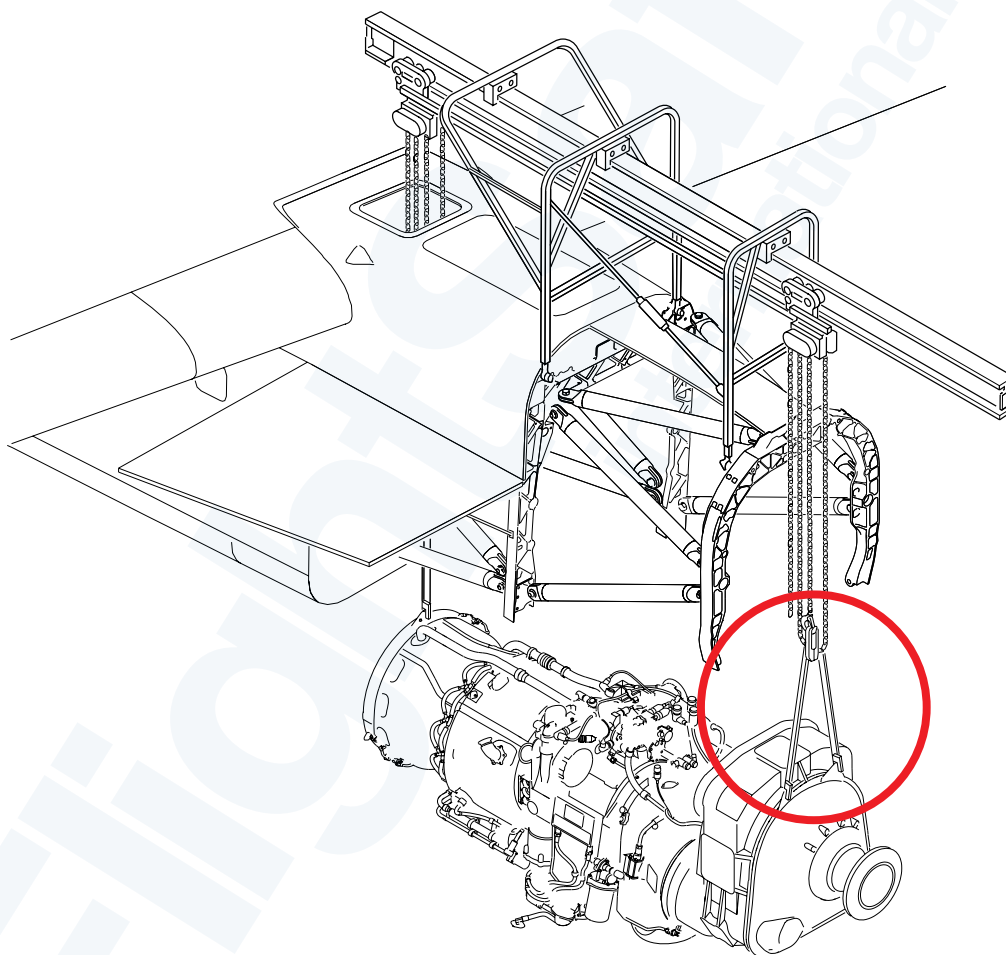
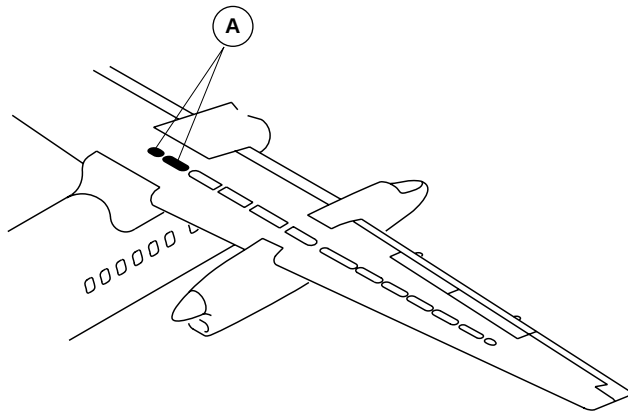
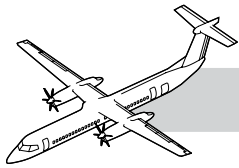


Figure 57-6. Hoist - Nacelle Mounted Engine

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LEGEND

- 1. Screw
- 2. Gasket.

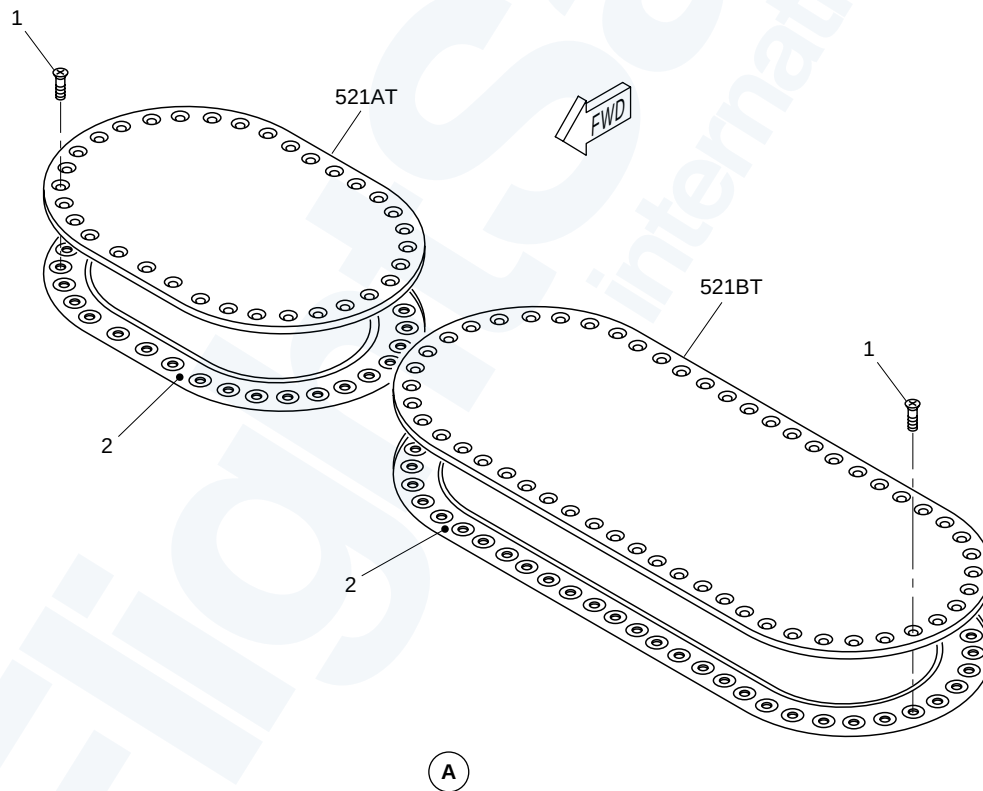
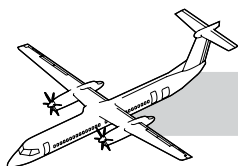


Figure 57-7. Dry Bay Fuel Access Panels (1 of 2)

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Center Wing Access Panels, Dry Bay

NOTE

Refer to:

- Figure 57-7. Dry Bay Fuel Access Panels (1 of 2).
- Figure 57-8. Dry Bay Fuel Access Panels (2 of 2).

Before you remove a wing access panel in the dry bay, you must perform the following procedure:

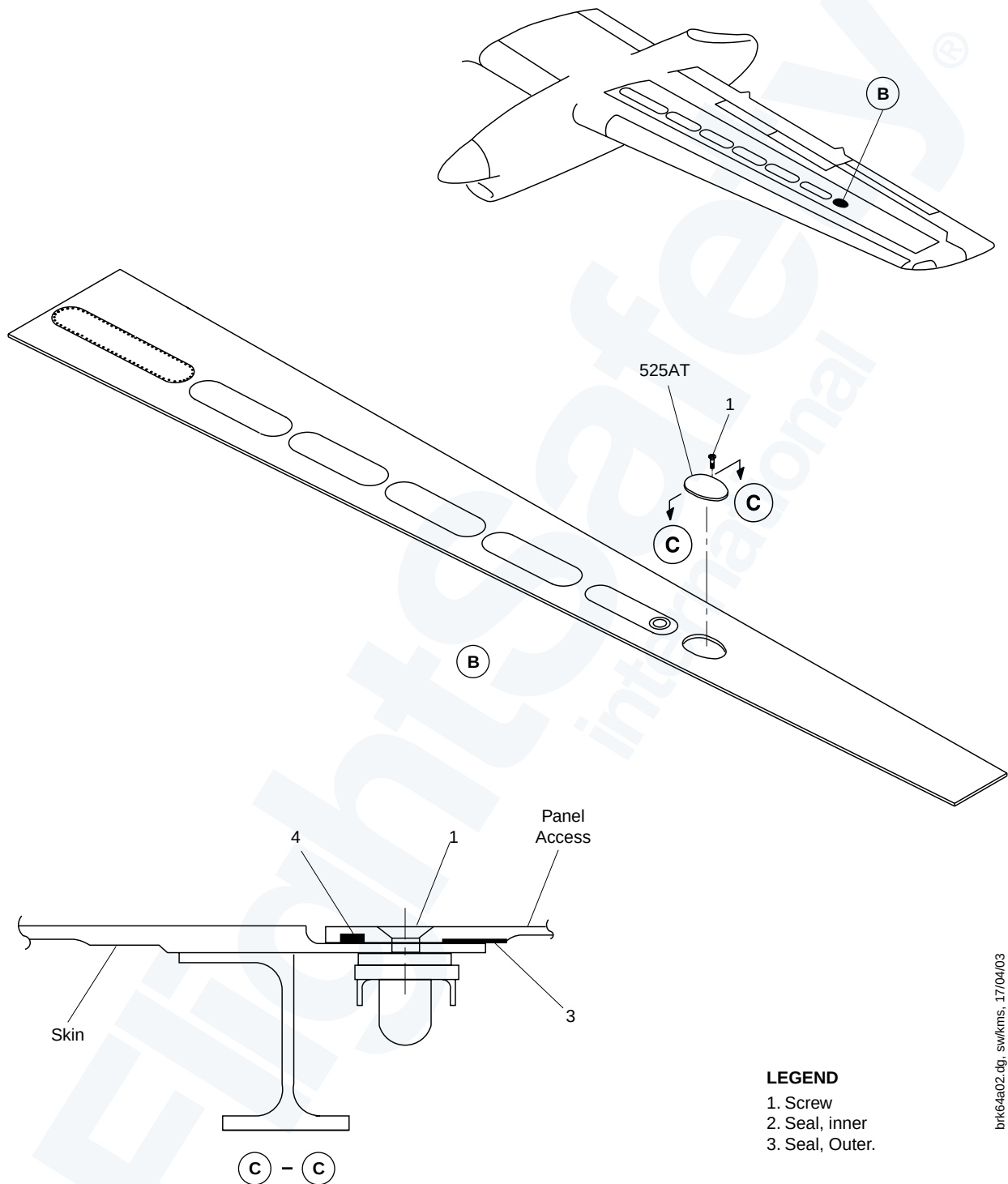
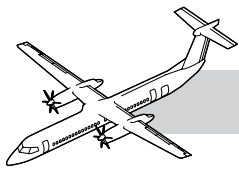
- Make sure that the aircraft is grounded
- On the DC CONTROL panel, make sure that all battery switches are in the OFF position
- Remove all the fuel from the tanks
- Do not move the aircraft.

NOTE

No de-fuelling is necessary when you do not remove more than one dry bay access panel.

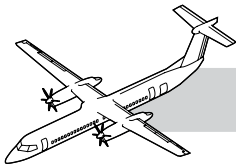
The dry bay center wing access panels are installed with screws.

The panels are made of machined aluminium alloy plate 7150-T7751. Gaskets with metallic washers are used in this location.



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Figure 57-8. Dry Bay Fuel Access Panels (2 of 2)



57-12-00 CENTER WING FAIRINGS

Refer to Figure 57-9. Center Wing Fairings.

These are the fairings on the external surfaces of the center wing that form an aerodynamic transition from the nacelle and fuselage to the inboard flap.

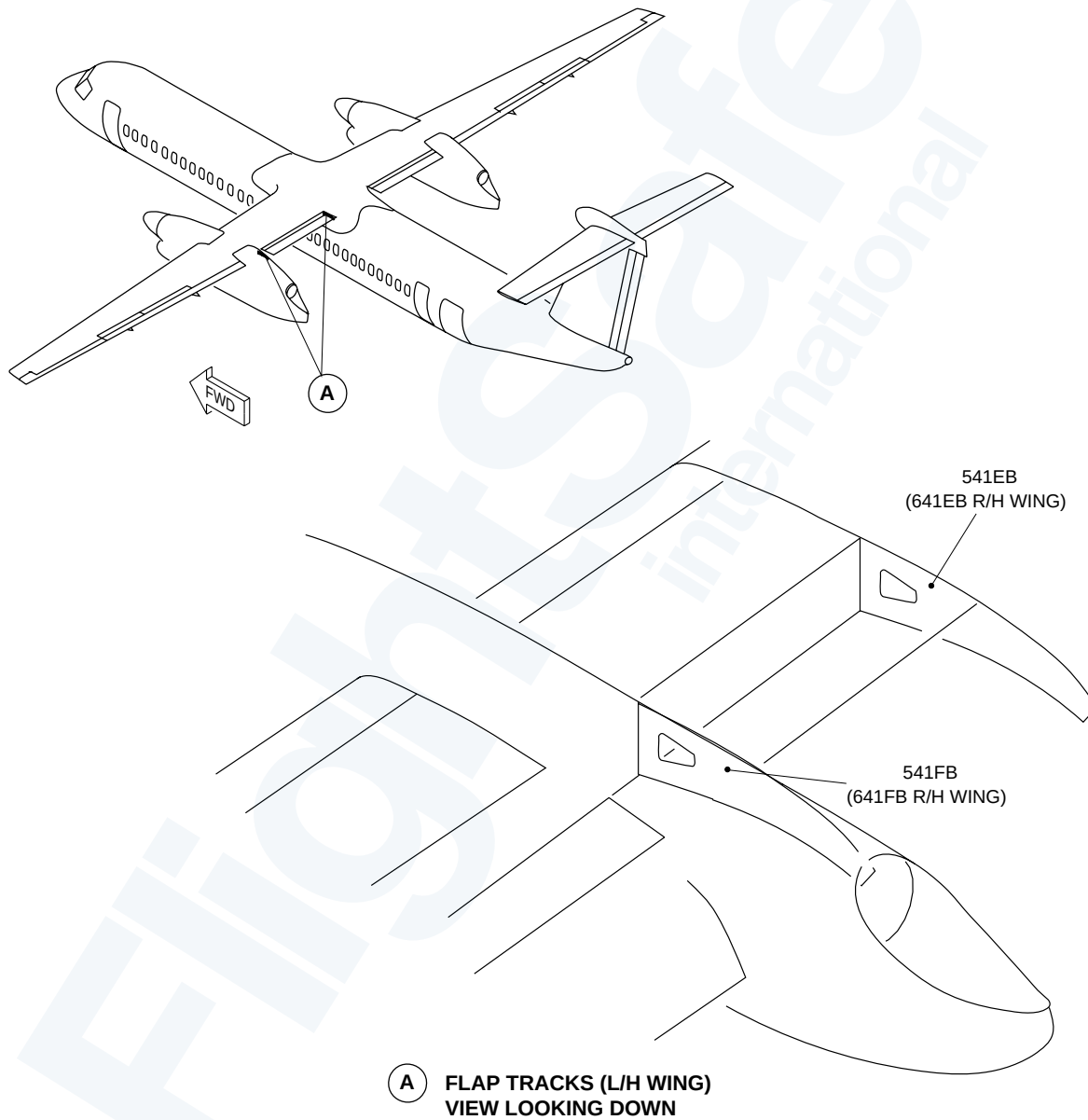
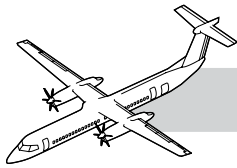


Figure 57-9. Center Wing Fairings



57-17-21 LANDING STRAPS

Refer to Figure 57-10. Landing Strap
- Removal/Installation.

The landing straps located on the upper wing surface provide aerodynamics between the nacelle and the wing mating edges.

57 WINGS

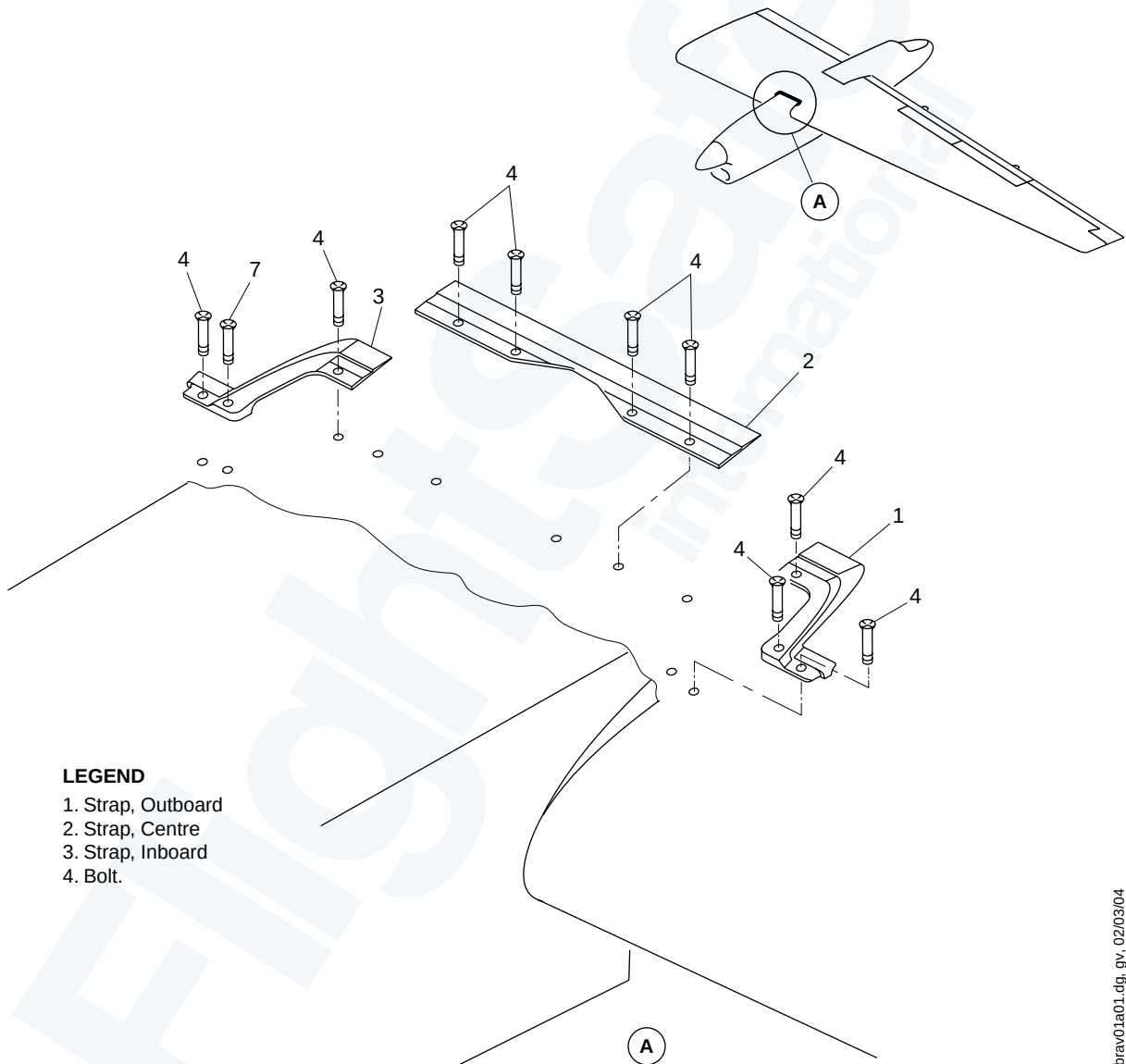
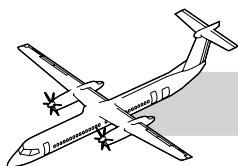


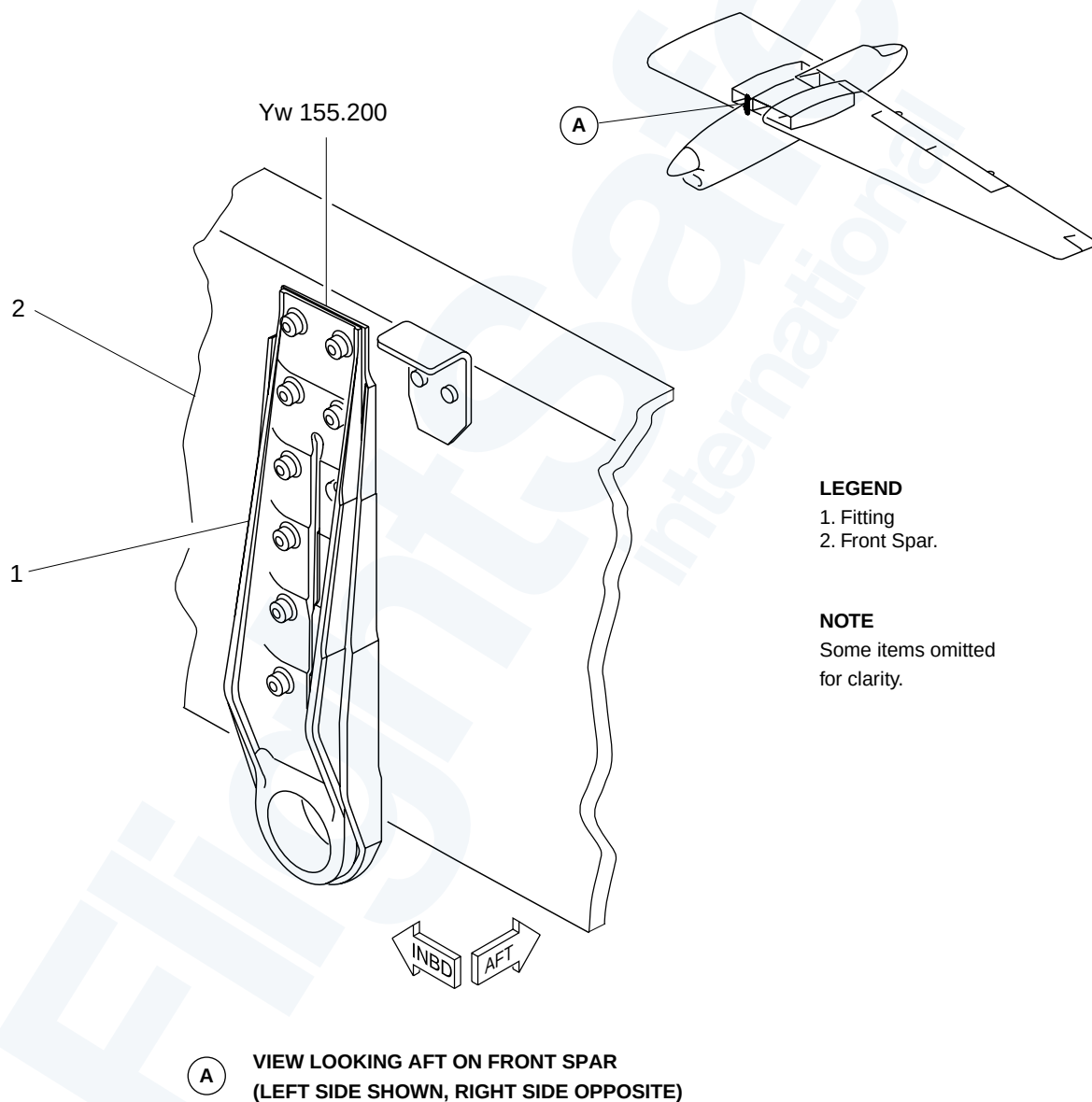
Figure 57-10. Landing Strap - Removal/Installation



57-18-00 NACELLE TO WING JOINT AT FRONT SPAR

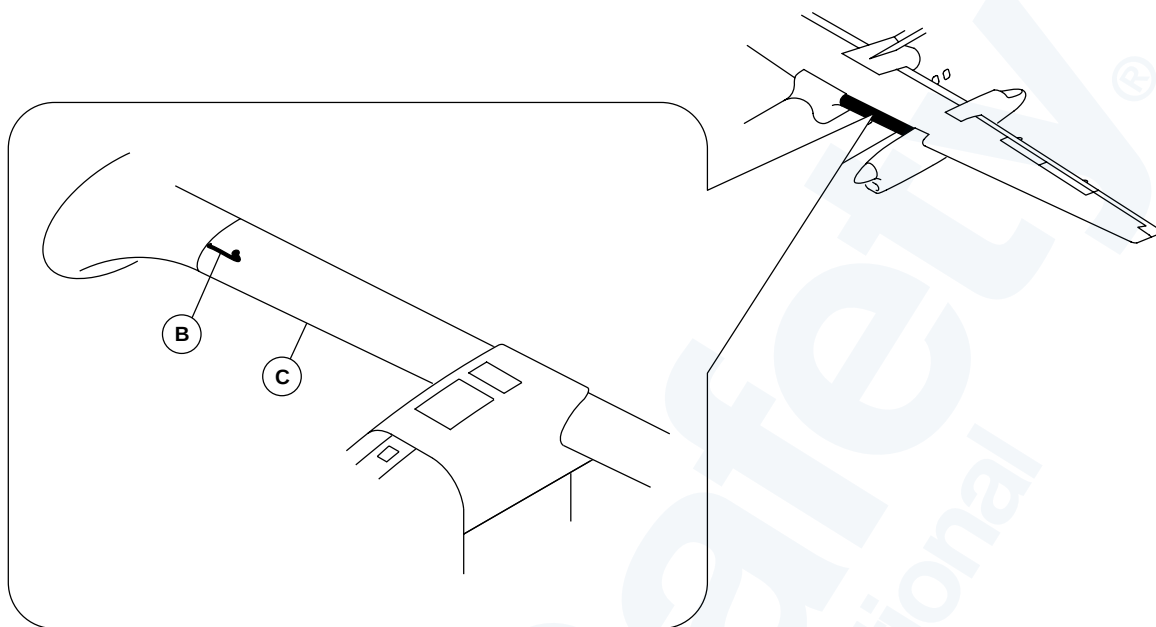
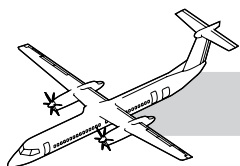
Refer to Figure 57-11. DVI of the Nacelle to Wing Joint at Front Spar, Sta Yw 155.2 LH and RH.

The two nested nacelle fittings on the front spar form the nacelle to wing joint at the front spar.



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Figure 57-11. DVI of the Nacelle to Wing Joint at Front Spar, Sta Yw 155.2 LH and RH



LEGEND

- 1. Screw
- 2. Leading edge
- 3. De-ice hose
- 4. Clamp
- 5. Silicone seal
- 6. De-icing line

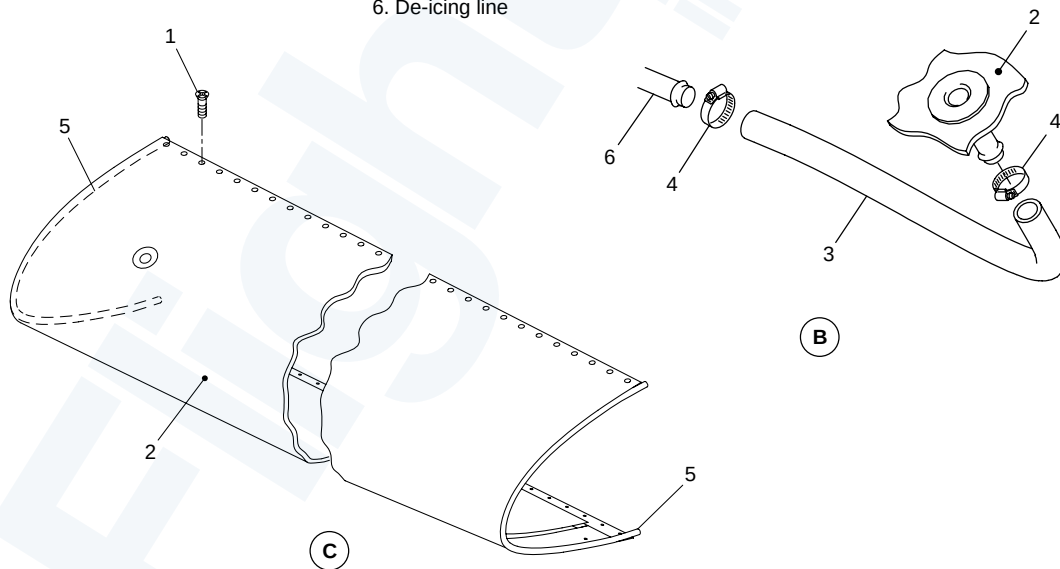
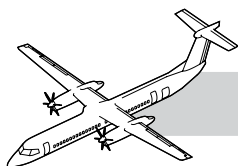


Figure 57-12. Center Wing Leading Edge



Center Wing Leading Edge

NOTE

Refer to Figure 57-12. Center Wing Leading Edge.

The center wing leading edge is installed with screws.

The center wing leading edge has an aluminium inner and outer skin and a flex-core.

Safety Precautions

Obey the fuel safety precautions when you do work on the fuel system or a fuel system component. If you do not do this, you can cause injuries to persons and damage to the equipment.

You must remove the fuel from the fuel tank in an open area. If you do not do this, you can cause injuries to persons and damage to the equipment.

Remove the fumes before you remove a fuel system component. Fuel fumes are dangerous and can be flammable.

Make sure that the fuel tender and the fuel nozzle are electrically bonded to the aircraft before you defuel or refuel the aircraft. An electrostatic spark during refuelling/defueling can cause an explosion or a fire.

Ground the aircraft and the work stand before you do fuel system maintenance. An electrostatic spark can cause an explosion or a fire.

Drain the fuel tank before you remove the access panels. If you do not do this, you can cause damage to the wing structure.

Make sure that you do not move the aircraft when the access panels are not installed. This is necessary to prevent damage to the wing and the wing box structure.

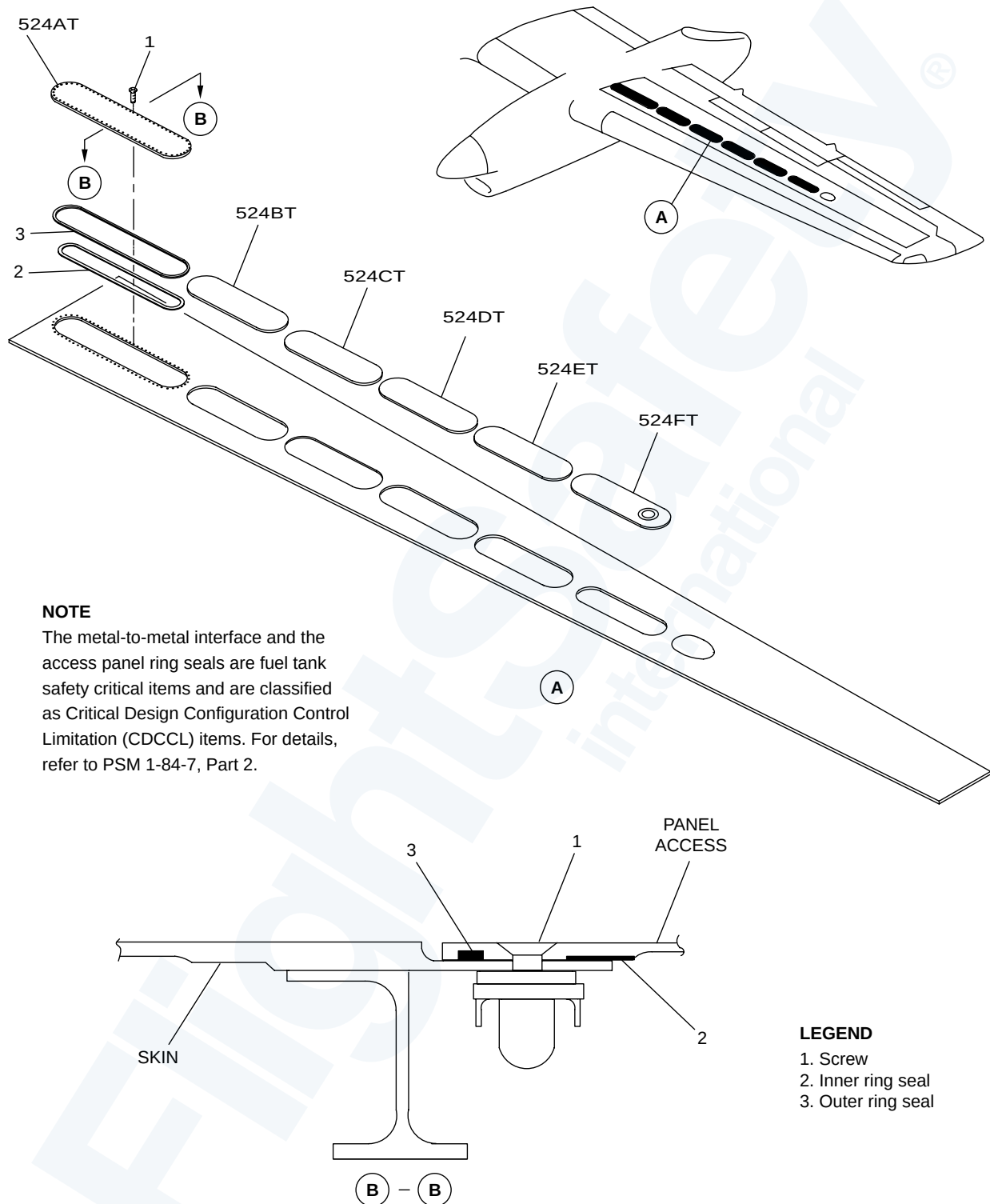
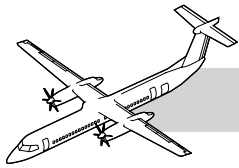
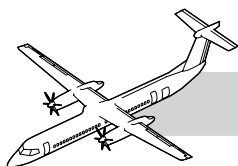


Figure 57-13. Outer Wing Fuel Access Panels



57-20-00 OUTER WING

INTRODUCTION

The left-hand outer wing stations are from Yw -192.142 to Yw -535.019.

GENERAL DESCRIPTION

The outer wing is an extension of the center wing and is a one-piece all metal unit mounted to the upper cabin of the fuselage through the center wing.

The left-hand outer wing box structure forms part of the integral fuel tanks to Station Yw -407.019.

Structural ladder plates located on the upper wing surface, give access to the outer wing integral fuel tanks.

The outer wing structure has the following components:

- Stringers
- Spar
- Ribs
- Outer wing box skin
- Access panels.

COMPONENT DESCRIPTION

Outer Wing Access Panels

The outer wing has access panels that follow:

- Outer Wing Fuel Access Panels.

Refer to:

- Figure 57-13. Outer Wing Fuel Access Panels.
- Figure 57-14. Outer Wing.

The outer wing fuel access panels are installed with screws.

The mating surfaces between the fuel tank access

panel and the wing skin is designed with metal-to-metal interface electrical bonding that must not be primed or painted. Only chemical conversion coating is permitted and other finishes are not permitted. The metal-to-metal interface and the access panel inner and outer ring seals (gaskets) are fuel tank safety critical items.

They are classified as a Critical Design Configuration Control Limitation (CDCCL) items. Their integrity must be maintained to make sure that unsafe conditions do not develop by maintenance, modification or repair. For the details, refer to PSM 1-84-7, Part 2.

- Geared tab linkage fairing, 532PB
- Aileron actuator fairing assembly, 543LB
- Spring tab fairing, 643MB (on right hand wing).

REMOVAL OF THE OUTER WING FUEL ACCESS PANELS

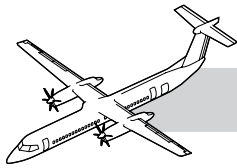
The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

The fuel access panels are on the top of the outer wing between the nacelle and the wing tip.

- Ground the aircraft and workstand.
- Do not damage the access panel inner and outer ring seals.
- Drain the fuel tank before you remove the access panels.

CAUTION

MAKE SURE THAT YOU DO NOT MOVE THE AIRCRAFT WHEN THE ACCESS PANELS ARE NOT INSTALLED. THIS IS NECESSARY TO PREVENT DAMAGE TO THE WING AND THE WING BOX STRUCTURE.

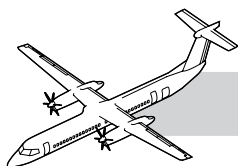


SAFETY PRECAUTIONS

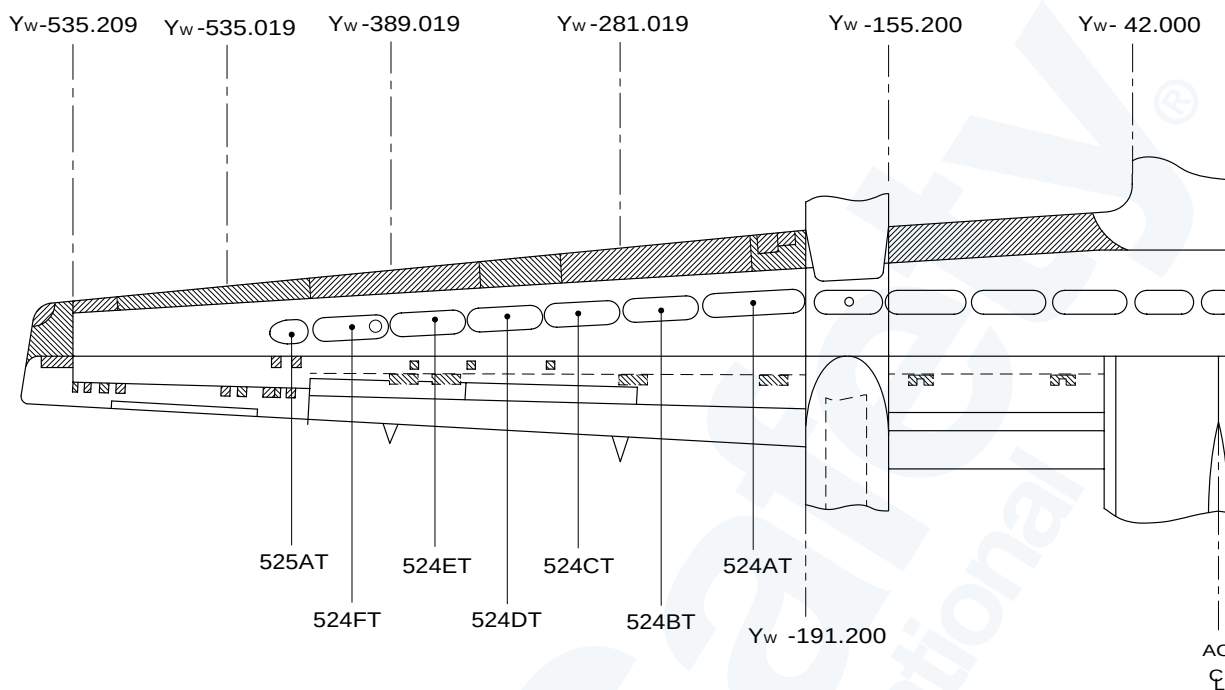
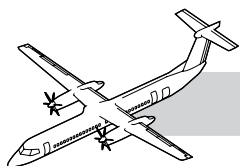
Do not paint the mating surface of the access panel and the wing skin. The metal-to-metal interface between the access panel and wing skin is necessary to provide electrical bonding. Painting of the mating surface will eliminate the electrical bonding. This can cause an unsafe condition and possibly can be an ignition source inside the fuel tank in the event of a lightning strike.

Do not damage the access panel inner and outer ring seals (gaskets). The gaskets are an isolator between the access panel and wing skin surface. The gasket prevents arcing between the access panel and the wing skin in the event of a lightning strike. Damage or modification to the gasket can cause an unsafe condition and possibly cause an ignition source in the fuel tank during a lightning strike.

Make sure that you do not move the aircraft before you install the fuel access panels. This is necessary to prevent damage to the wing and the wing box structure.



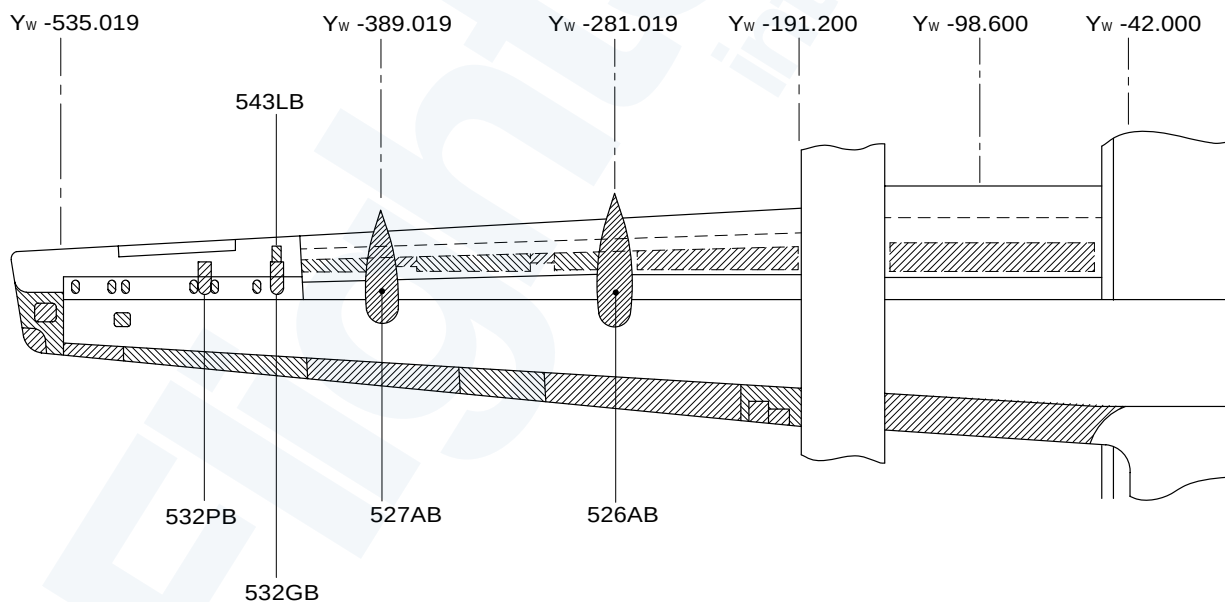
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OUTER WING ACCESS PANELS (L/H WING) VIEW LOOKING DOWN

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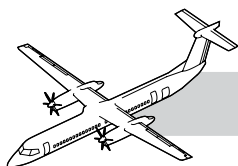
Figure 57-14. Outer Wing



OUTER WING FAIRINGS (L/H WING) VIEW LOOKING UP

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Figure 57-15. Outer Wing Fairings



Outer Wing Fairings

NOTES

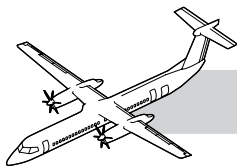
Refer to Figure 57-15. Outer Wing Fairings.

The outer wing fairings have the components that follow:

- Flap track #4 fixed fairing, 526AB
- Flap track #5 fixed fairing, 527AB
- Aileron actuator shroud fairing, 532GB
- Geared tab linkage fairing, 532PB
- Aileron actuator fairing assembly, 543LB
- Spring tab fairing, 643MB (on right hand wing).

SYSTEM <u>57</u> <u>WINGS</u>		1. NUMBER OF PARTS REQUIRED FOR ALL FLIGHT CONDITIONS EXCEPT AS PROVIDED IN COLUMN 2.	
ATA SUB-SYSTEM	PART DESCRIPTION	4	2. REMARKS AND/OR EXCEPTIONS
57-2	FLAP TRACK #4 and #5 AFT FAIRINGS		Any number of fairings may be missing with no performance penalty.

Figure 57-16. CDL - FLAP TRACK #4 and #5 AFT FAIRINGS



Outer Wing Leading Edge

Refer to Figure 57-17. Outer Wing Leading Edge.

The outer wing leading edge is installed with screws.

The center wing leading edge is made of cloth-aramid fabric plies with a honeycomb core.

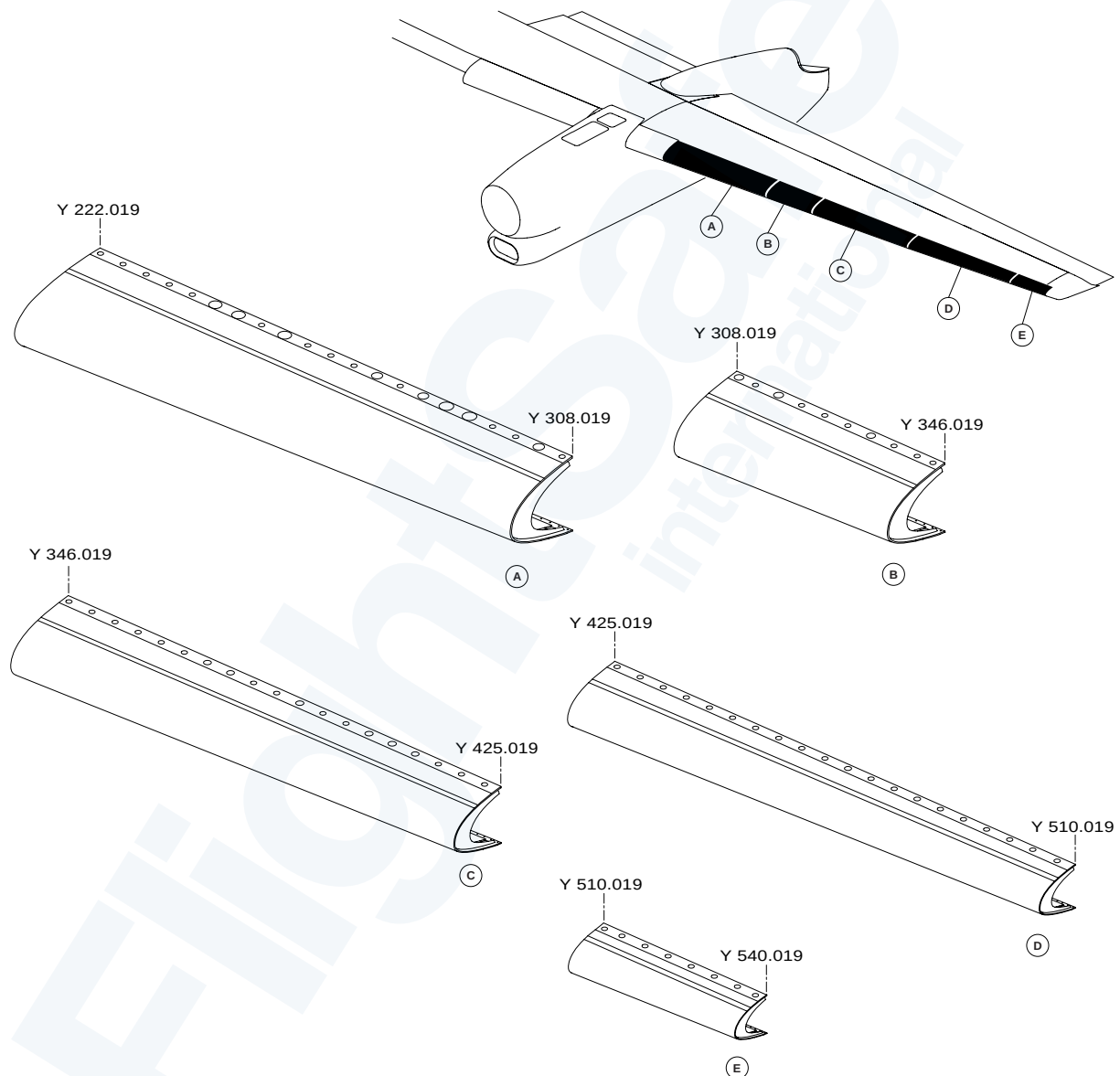
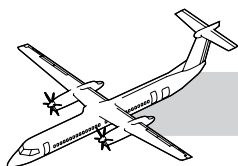


Figure 57-17. Outer Wing Leading Edge



Wing Tip Fairings

Refer to Figure 57-18. Wing Tip Fairings.

The wing tip fairing is installed at the end of the left and the right wing structure with screws.

The wing tip fairing is made of cloth-aramid and glass fabric plies with a honeycomb core.

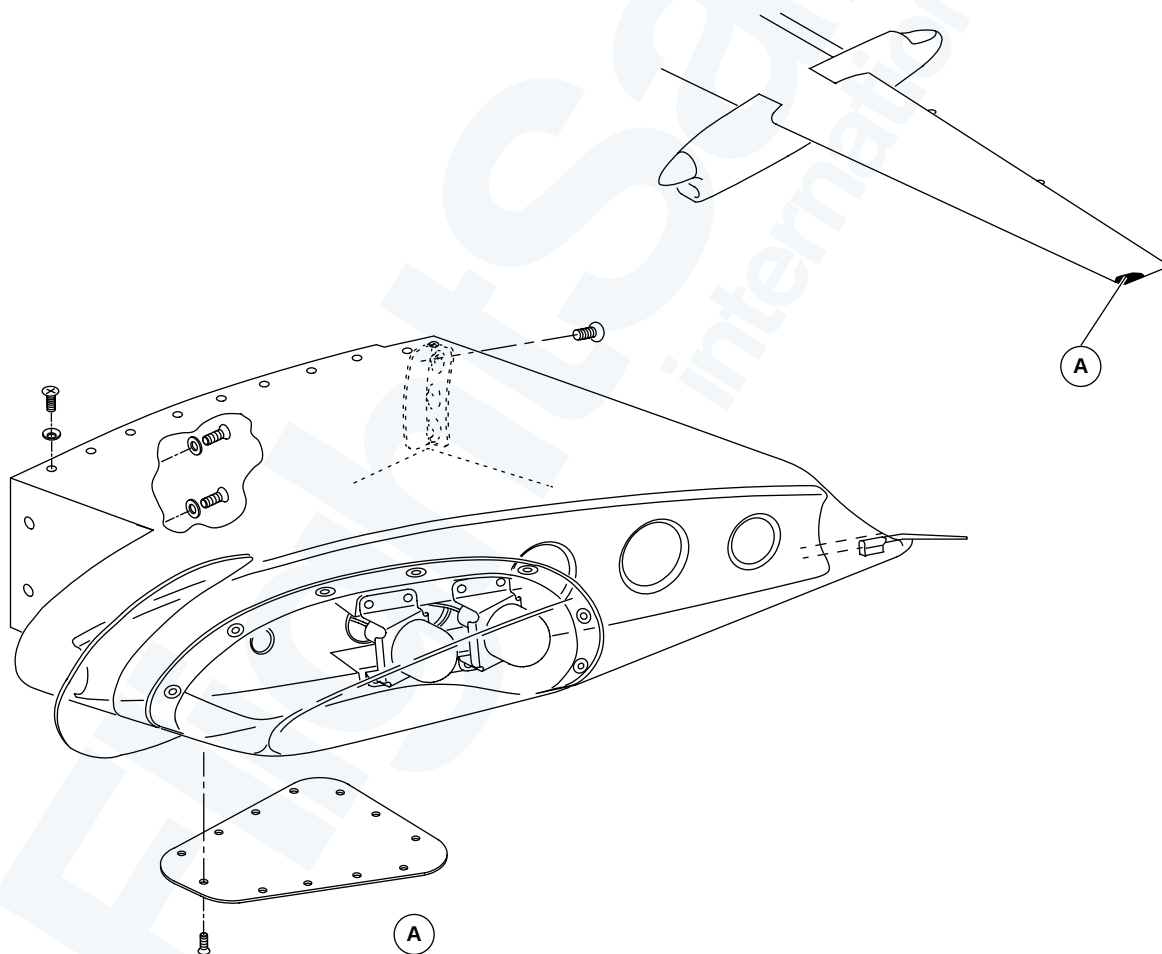
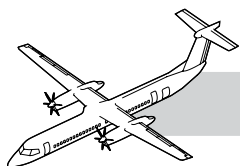


Figure 57-18. Wing Tip Fairings



LEGEND

1. Left outboard flap
2. Left inboard flap
3. Right inboard flap
4. Right outboard flap

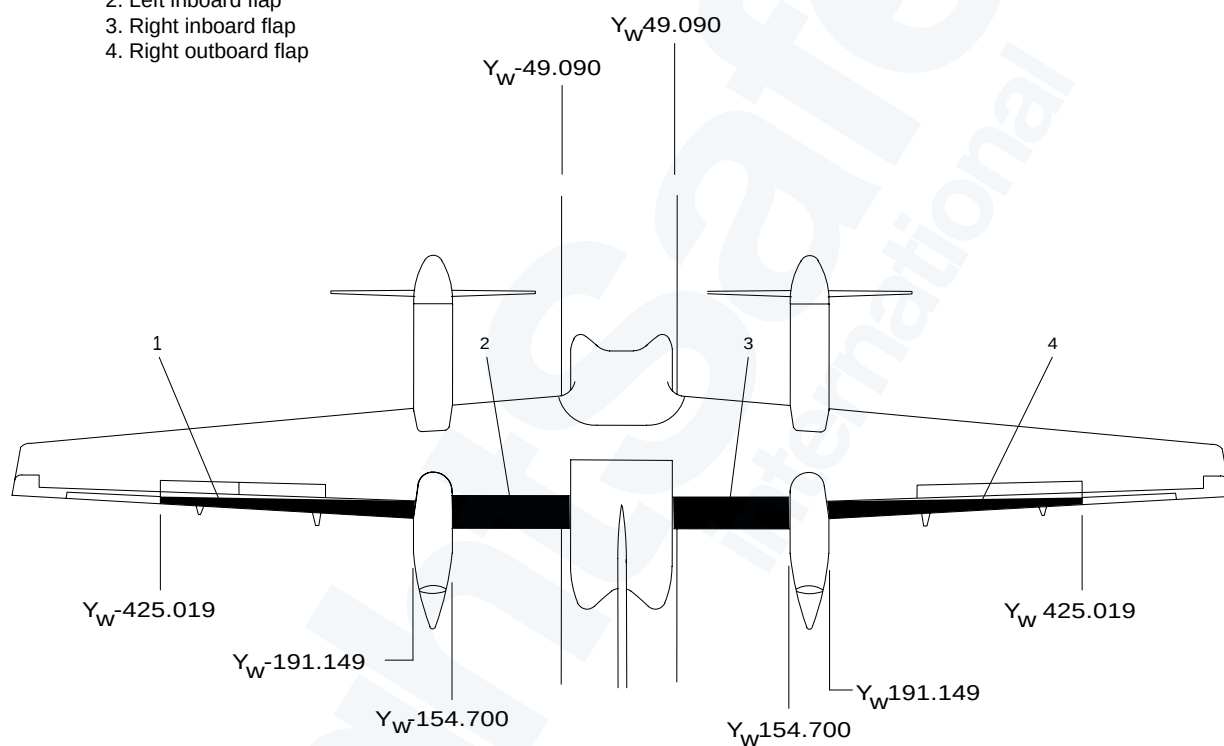
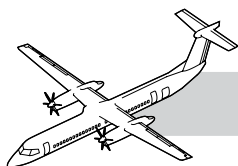


Figure 57-19. Wings Flaps



57-50-00 TRAILING EDGE DEVICES/FLAPS

NOTES

INTRODUCTION

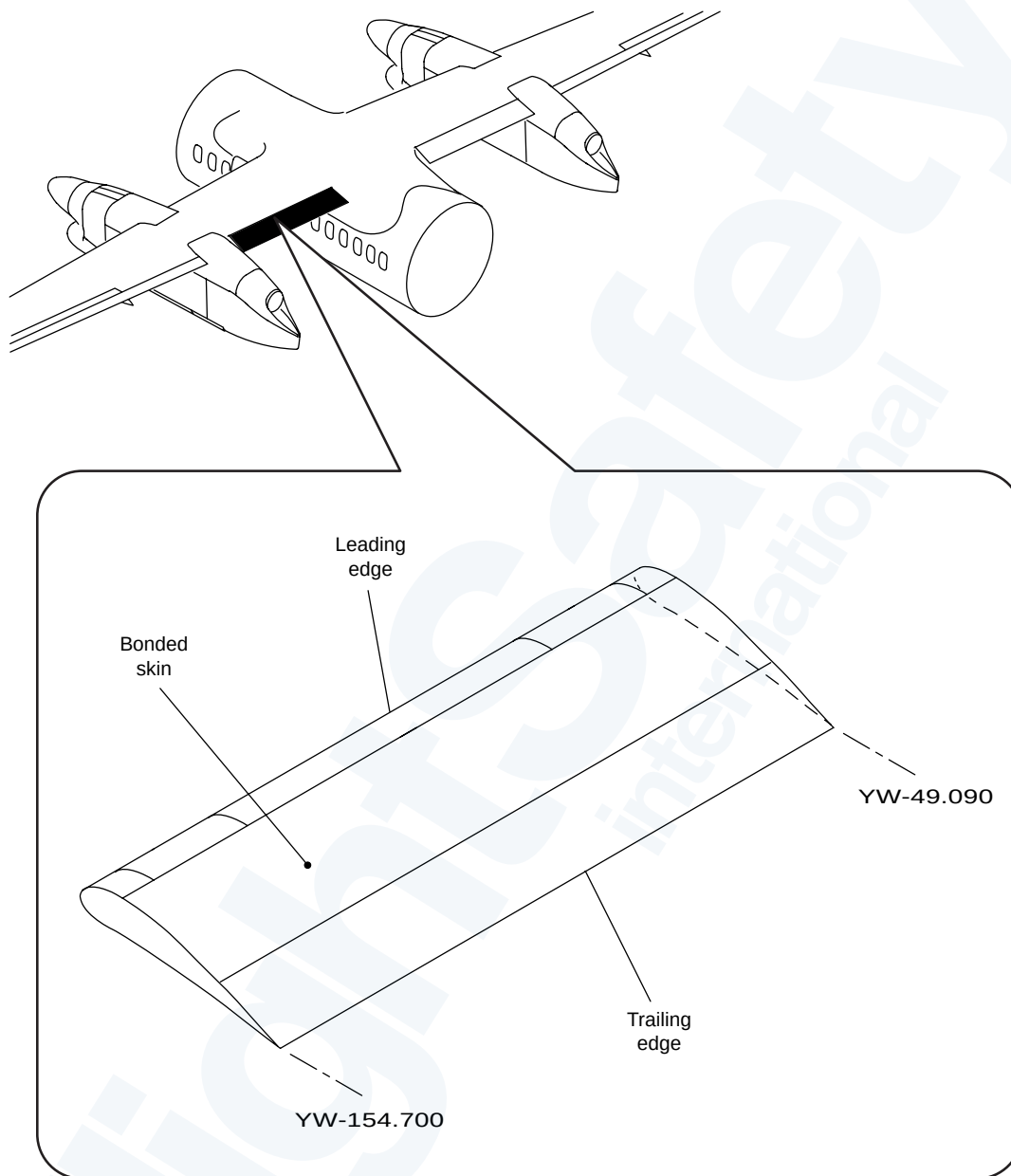
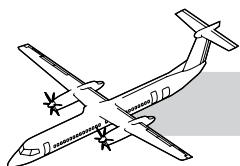
The inboard wing flaps are positioned between stations Yw -49.09 and Yw -154.70, and between Yw -191.419 and Yw -425.019 for outboard on the left hand side. (Right hand side is opposite).

GENERAL

Refer to Figure 57-19. Wings Flaps.

The inboard and outboard flaps have the following components:

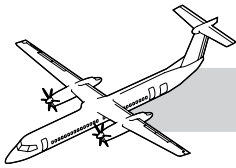
- Flap leading edge
- Flap trailing edge
- Stringers
- Front and rear spars
- Ribs
- Attached fittings
- Flap skin.



NOTE

LH is shown, RH is opposite.

Figure 57-20. Inboard Flaps



COMPONENT DESCRIPTION

NOTES

Inboard Flap

Refer to:

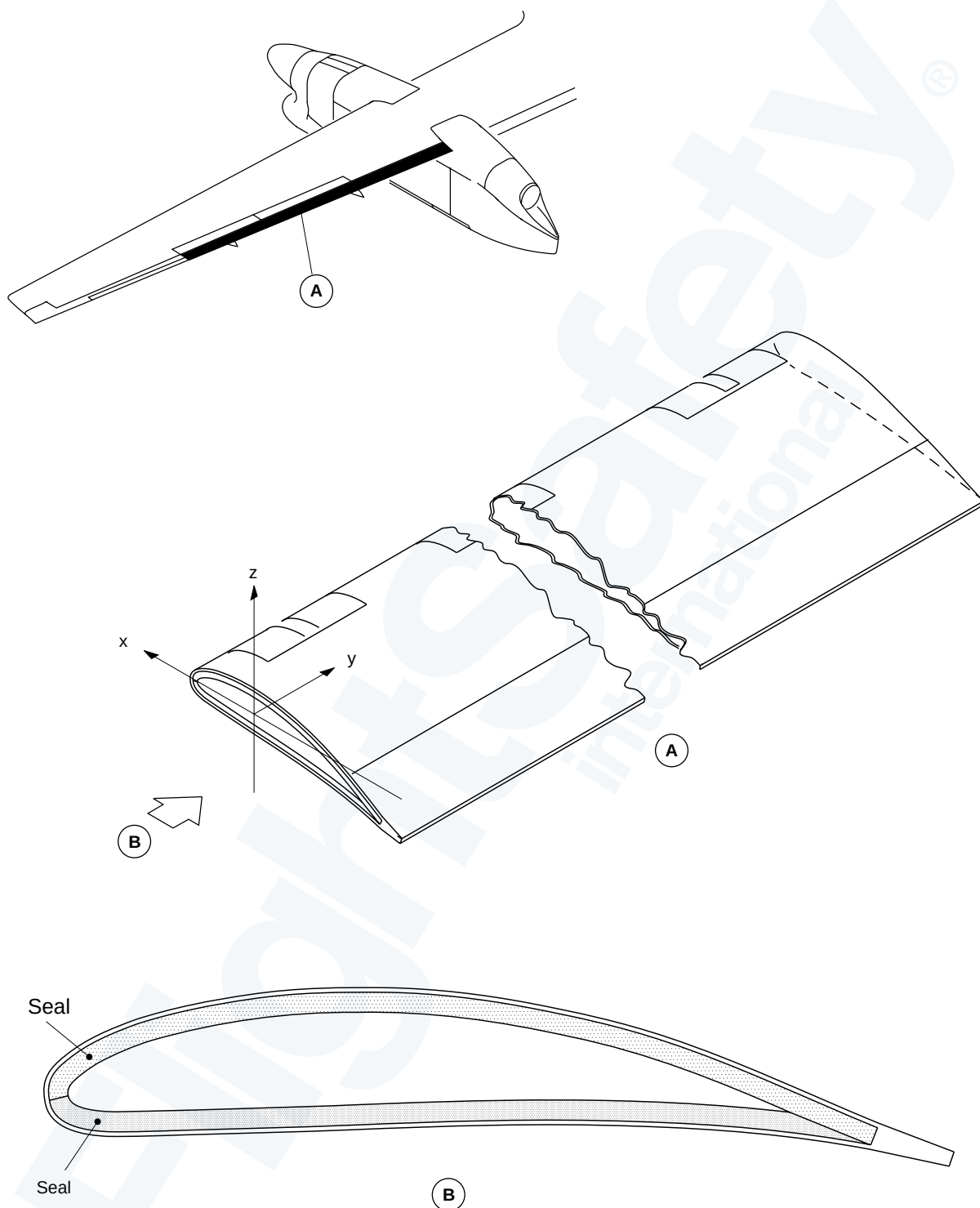
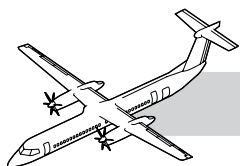
- Figure 57-20. Inboard Flaps.
- Figure 57-21. Flap LH Inner Tape.

The main box frame is made of aluminium alloy, using front and rear spars, chord wise ribs, span wise stringers and bonded skin.

The leading and trailing edges are made of composite materials and use chord wise stiffeners for strength.

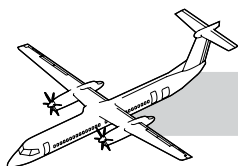


Figure 57-21. Flap LH Inner Tape



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Figure 57-22. Visual Inspection of the Outboard Flap Seal



Outboard Flaps

NOTES

Refer to Figure 57-22. Visual Inspection of the Outboard Flap Seal.

The outboard and inboard flaps have similar construction.

57-50-06 OUTBOARD FLAP SEAL

Refer to Figure 57-23. Flap Outer.

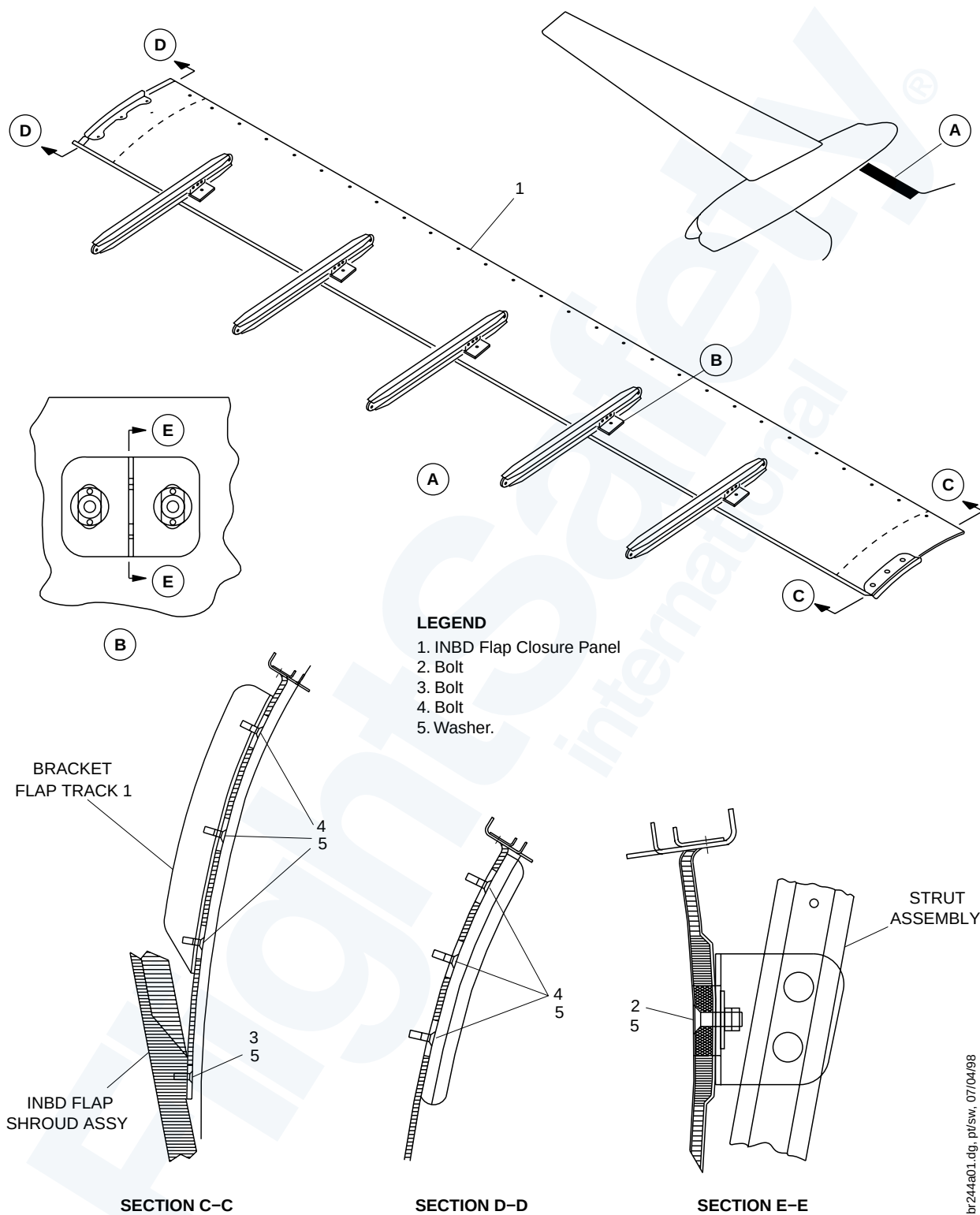
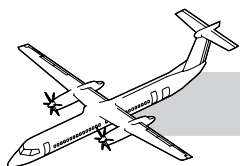
Visually inspect the outboard flap seals in accordance with SRM TOPIC 57-52-10-01-02 Damage Limits for the Seals.

Examine the flap seals for cuts, or missing material.

If you find any cuts or missing material, Refer to SRM TOPIC 57-52-10-01-02 for the allowable damage limits.

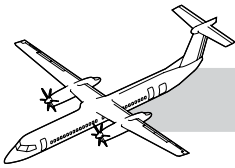


Figure 57-23. Flap Outer



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Figure 57-24. Inboard Flap Closure Panel



57-1-01/06 INBOARD AND OUTBOARD FLAP CLOSURE PANELS

NOTES

Refer to:

- Figure 57-24. Inboard Flap Closure Panel.
- Figure 57-25. Outboard Flap Closure Panels.

The closure panel attaches to the upper flap shroud strut assembly.

In the fully retracted position, the leading edge of the flap aligns with the upper flap shroud on the wing to give an aerodynamically smooth wing surface. The flaps must be lowered to access the flap closure panels.

They can be removed to gain access to the components that follow:

- Flap actuators
- Ballscrew jacks
- Flap tracks
- Flap transmission shafts
- Hydraulic lines
- Electrical harnesses.

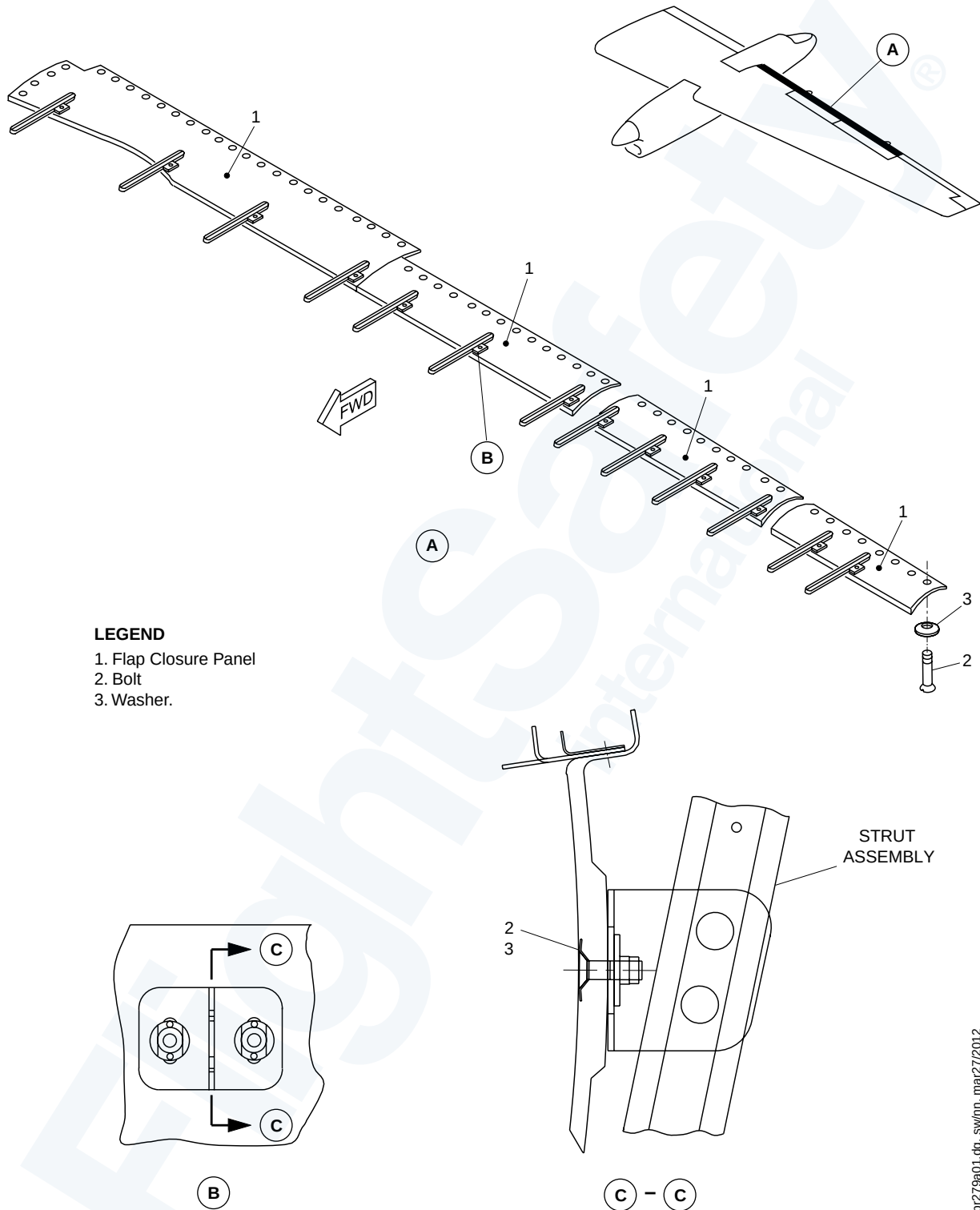
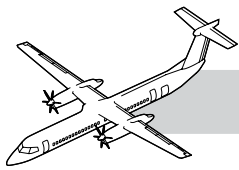
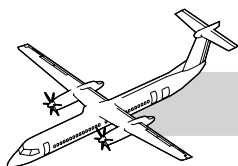
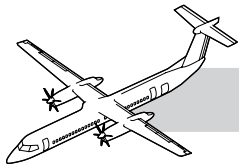


Figure 57-25. Outboard Flap Closure Panels

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LEGEND

1. Right aileron
2. Ground adjustable trim tab
3. Aileron geared tab
4. Aileron geared tab
5. Left aileron

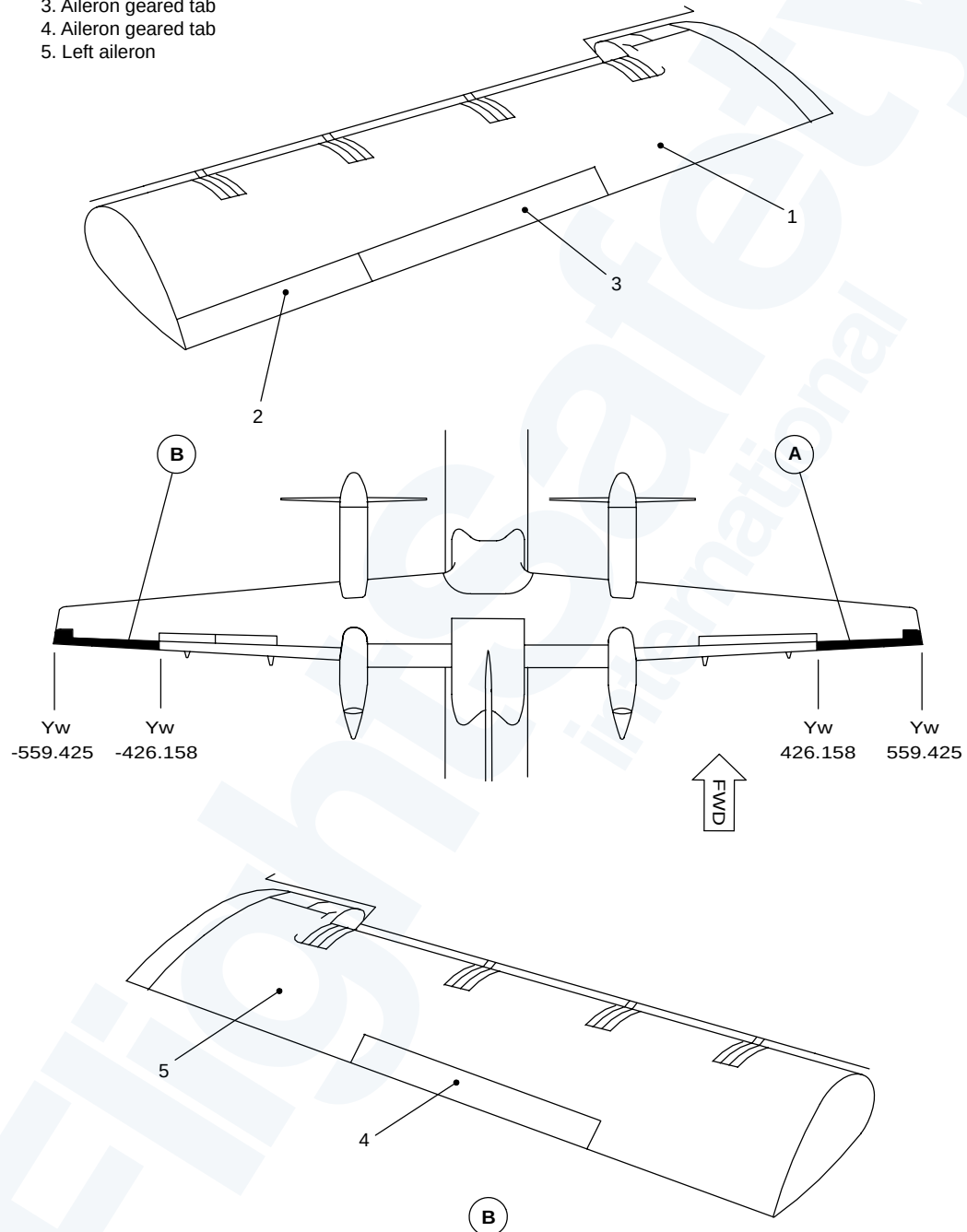
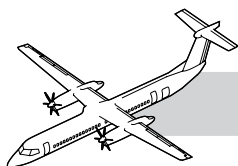


Figure 57-26. Wing Flaps



57-60-00 AILERON

NOTES

INTRODUCTION

The ailerons are positioned between station Yw -426.158 and Yw -554.000 for the left hand side. (Right hand side is opposite).

GENERAL

Refer to Figure 57-26. Wing Flaps.

The aileron is made of aluminum alloy using front and rear spars, stiffeners and bonded skins.

The ailerons have the components that follow:

- Aileron trailing edge
- Front and rear spars
- Stiffeners
- Attached fittings
- Aileron tip assembly
- Aileron skin.

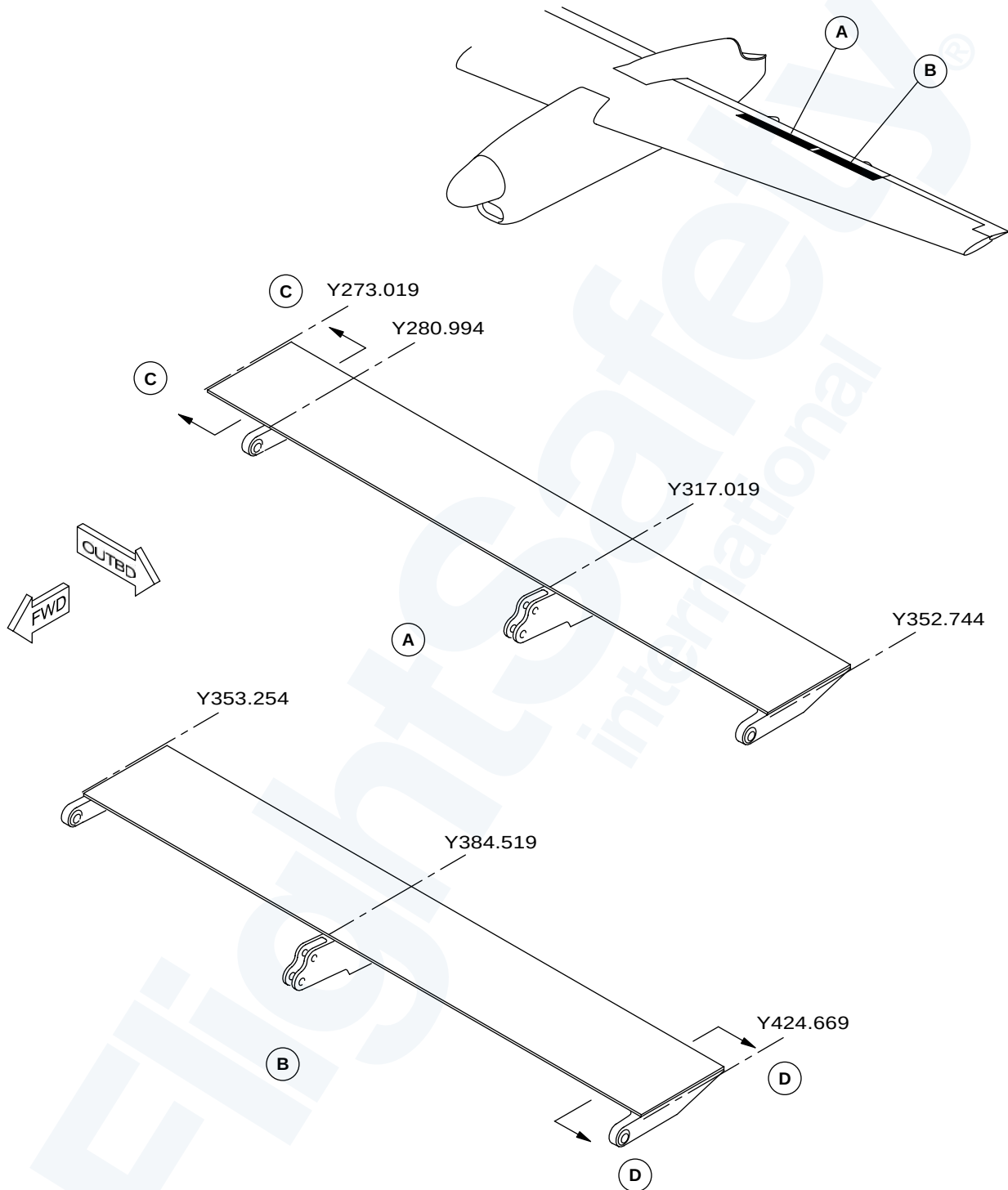
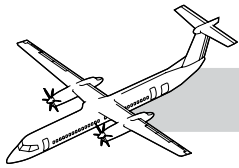
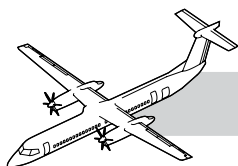


Figure 57-27. Inboard and Outboard Spoilers (1 of 2)



57-70-00 SPOILERS

NOTES

INTRODUCTION

The spoilers are positioned between station Yw -273.019 and Yw -425.019 for the left hand side. (Right hand side is opposite).

GENERAL

Refer to Figure 57-27. Inboard and Outboard Spoilers (1 of 2).

The general structural arrangement of the spoiler are an aluminium upper and lower skin bonded to an aluminium honeycomb core with bonded ribs.

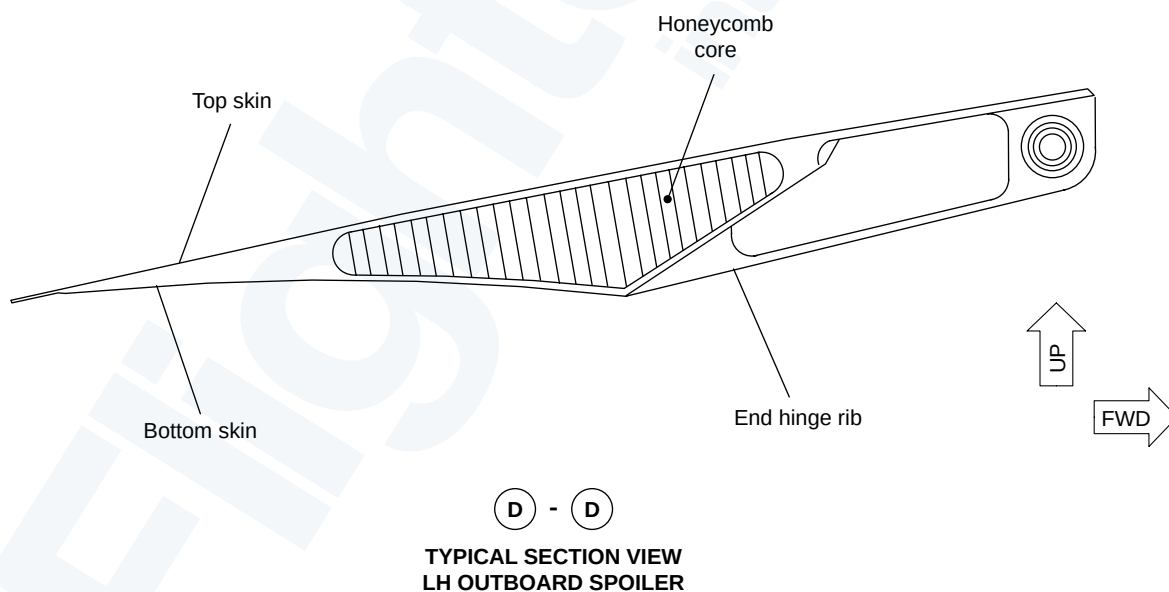
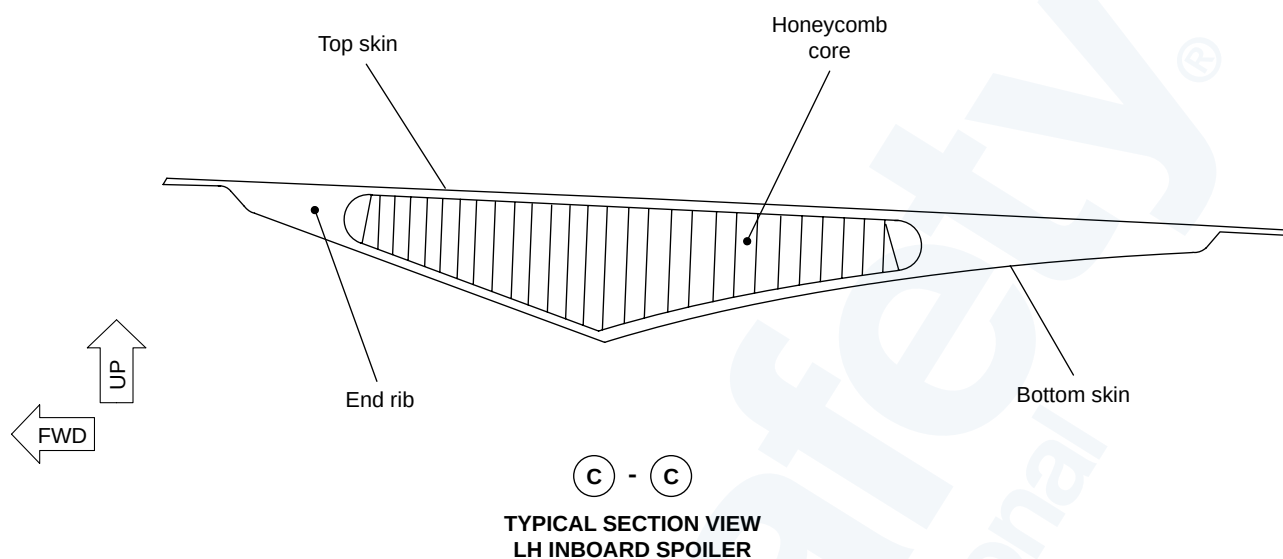
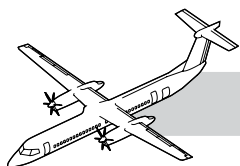
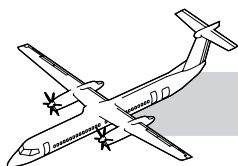


Figure 57-28. Inboard and Outboard Spoilers (2 of 2)



COMPONENT DESCRIPTION

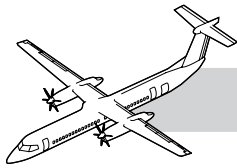
NOTES

Inboard and Outboard Spoilers

Refer to Figure 57-28. Inboard and Outboard Spoilers (2 of 2).

The spoilers have the following components:

- Top skin
- Bottom skin
- Honeycomb core
- End hinge rib, station Yw -353.254
- Rib, station Yn -384.519
- Main hinge assembly, station Yw -384.519
- End hinge rib, station Yn -424.669.



57-00-00 APPENDIX

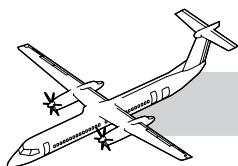
NOTES

MAINTENANCE CONSIDERATION

Unscheduled Inspection

Refer to the Bombardier published AMM Part 2 PSM 1-84-2.

- TASK 05-50-06-210-801
Inspection after a Hard Landing
- TASK 05-50-11-210-801
Inspection after Extreme Turbulence or Buffeting
- TASK 05-50-16-210-801
Inspection after a Bird Strike Condition
- TASK 05-50-21-210-801
Inspection after a Lightning Strike
- TASK 05-50-21-210-802
Inspection after a Lightning Strike for a Ferry Flight Only



57-00-00 SPECIAL TOOL & TEST EQUIPMENT

- Commercially Available Calibration Test Standard for HFEC inspection, P/N SRS 0824A/7075T6 (NDT Engineering Corporation)
- Commercially Available Calibration Test Standard for HFEC inspection, P/N SRS 0824S (NDT Engineering Corporation)
- GSB0700007 Jack Adapters
- GSB0700019 Wing Jacks
- GSB0700020 Front Fuselage Jack
- GSB0700025 (or equivalent) Engine strap
- GSB0700024 Stand - Tail Support
- Commercially available Load cell (able to measure a 2500 lb (1133.98 kg) load)
- GSB2000001 Borescope - 115volt, 60Hz
- GSB2000015 Borescope - 220volt, 50Hz
- Commercially Available Calibration Test Standard for LFEC inspection, P/N DHC 29445-001
- Commercially Available from Bombardier Aerospace (Toronto) Calibration Test Standard for LFEC inspection, P/N DHC 29445-001
- Available for rent from Bombardier Aerospace (Toronto) Calibration Test Standard for Ultrasonic inspection, P/N EXBD 400-50040
- Available from Bombardier Aerospace (Toronto) Calibration Test Standard (CTS), P/N EXBD400-50047-001
- Commercially Available 25X Torque Multiplier
- GSB2700008 Rigging Pin Kit